

Agentic AI for Maritime Freight Pricing and Route Optimization

FreightAI Platform: Complete Technical Architecture, Autonomous AI Agents, 5-Factor MCDA Risk Gating, and Governed Carrier Verification State Machine.

Lead Engineer: Naresh Ramavath	Project Title: Agentic AI for Maritime Freight Pricing and Route Optimization	Program: Infosys Springboard Virtual Internship 7.0
Team / Cohort: Group 1	Duration: 3 Months (July 2026 – September 2026)	Verification Status: Production Ready (63/63 Tests Passed)
Frontend Stack: React 18, Vite 8, Tailwind CSS, Leaflet, Chart.js	Backend Stack: Django 5.x REST Framework, Python 3.11+, Scikit-Learn	Deployment: Vercel (Client) + Cloudflare Secured API Tunnel

1. Executive Summary & Problem Formulation

Commercial international freight forwarding operations have historically relied on fragmented manual spreadsheets, email-based spot rate inquiries, and disconnected compliance checks. Standard quote turnaround times range from **24 to 48 hours**, resulting in lost bookings, opaque pricing, and high exposure to unexpected transit delays, customs holds, and port demurrage costs. Furthermore, once quote options are generated, the manual coordination between shippers and carriers for capacity confirmation, rate adjustments, and customs sign-off introduces major friction.

FreightAI resolves this systemic problem by deploying a unified, end-to-end multi-agent logistics platform. The system ingests cargo specifications and international trade parameters, predicts competitive multi-carrier spot rates using machine learning, evaluates marine weather hazards and customs compliance in parallel, gates risks via a 5-factor Multi-Criteria Decision Analysis (MCDA) engine, and orchestrates a secure, multi-tenant Milestone 4 booking confirmation workflow in **under 7 seconds**.

Milestone	Core Focus & Functional Scope	Key Architectural Deliverables	Operational Outcome
Milestone 1	Core Shipment Foundation, Geospatial Routing & Baseline Rule Pricing	UN/LOCODE port geocoding, Great-Circle route geometry, chargeable weight resolver, Incoterms 2020 logic, itemized base pricing.	Standardized shipment request intake & deterministic baseline quote.
Milestone 2	AI/ML Predictive Pricing Intelligence & Margin Optimization	Gradient Boosted Regressor (GBR) trained on 5,000 historical trade lanes (R²=0.874), Gini feature explainability, multi-carrier rate cards.	Market-driven spot rate predictions for Maersk, MSC, CMA CGM, Hapag-Lloyd.
Milestone 3	Parallel AI Services & 5-Factor MCDA Composite Risk Engine	Live Open-Meteo marine weather telemetry, Customs Hybrid RAG (BM25 + TF-IDF) over 10 trade jurisdictions, 5-factor risk scoring.	Automated risk gating: Low Risk (Instant Quotes) vs High Risk (Customs Desk).
Milestone 4	Carrier Selection, Role-Based Verification & Governed Booking State Machine	Multi-company isolation, carrier verification desk, counter-offer negotiation, digital customs release, immutable booking confirmation.	End-to-end commercial transaction closure with legally valid booking reference.

2. Autonomous AI Agent Architecture & End-to-End System Flow

The FreightAI architecture cleanly decouples frontend stakeholder interfaces from backend intelligent services. When a quotation request is initiated, the **AI Orchestrator** fans out sub-tasks concurrently across specialized agents, synthesizes their outputs, and persists the resulting state machine snapshot:

Service / Agent	Core Responsibility	Inputs / Telemetry Sources	Generated Outputs
AI Orchestrator	Manages asynchronous agent concurrency, merges sub-task responses, executes the 5-factor risk gate, and writes state transitions.	Shipper payload, auth session tokens	Consolidated shipment quote, verification ID
Route Engine	Calculates geodesic distance, navigational waypoints, chokepoint canal feasibility, and determines accurate transit ETAs.	UN/LOCODE port pairs, bathymetric data	Nautical distance (NM), transit days, waypoints
ML Pricing Agent	Executes Gradient Boosted regression against 5,000 lane benchmarks to calculate dynamic market spot rates with profit margin floors.	Weight, volume, lane distance, fuel index	Predicted spot rates, carrier options, margin %
Weather Agent	Polls live oceanographic telemetry at route coordinates via Open-Meteo API. Identifies tropical storms, high swell, and gale-force winds.	Open-Meteo Marine API, coordinate cache	Wave height, swell period, weather risk score
Customs RAG Agent	Performs lexical (BM25) and semantic (TF-IDF) vector retrieval across 10 international customs jurisdictions to audit required trade forms.	6-digit HS codes, country trade regulations	Mandatory doc checklist, tariff flags, hold risk
Risk Engine (MCDA)	Aggregates 5 weighted risk vectors into a composite score (0–100) and executes statutory policy routing.	Weather, customs, delay, OTP, geopolitics	Composite risk index, routing gate decision

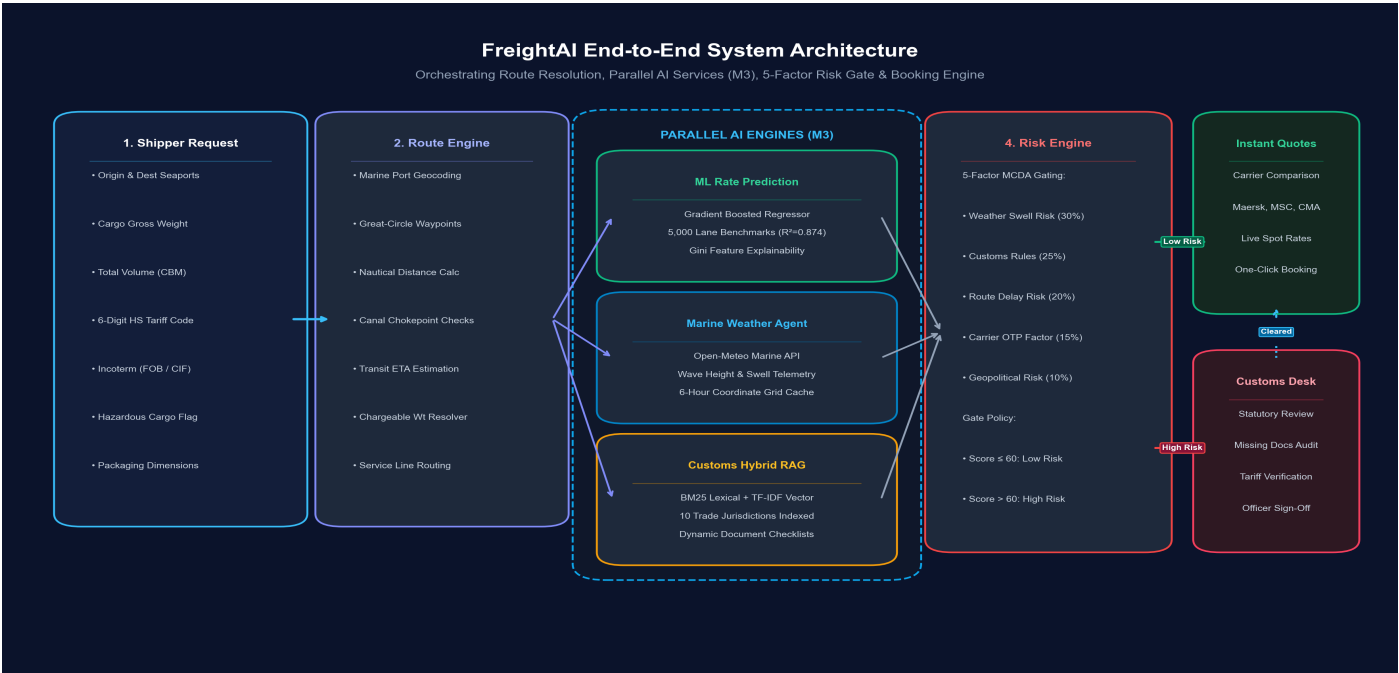


Figure 1: FreightAI End-to-End System Architecture — Input intake, parallel AI execution, 5-factor risk gate, and decision routing.

3. Milestone 1: Shipment Intake, Route Engine & Baseline Pricing

Milestone 1 establishes the deterministic foundation of FreightAI. It processes cargo consignments, computes nautical geography, enforces Incoterms commercial boundaries, and computes rule-based baseline rate structures:

- **Chargeable Weight Resolution:** Implements the international maritime volumetric standard: Chargeable Weight (kg) = max(Gross Cargo Weight, Total Volume in CBM × 1,000 kg). This prevents carrier revenue leakage on low-density, bulky consignments.
- **Great-Circle Nautical Route Engine:** Calculates spherical geodesic trajectories between international seaports using the Haversine trigonometric formulation: $d = 2R \cdot \arcsin(\sqrt{\sin^2(\Delta_{lat}/2) + \cos(lat_1) \cdot \cos(lat_2) \cdot \sin^2(\Delta_{lon}/2)})$, adjusted with a 12% oceanic routing detour coefficient to account for navigation channels.
- **Maritime Chokepoint Detection:** Automatically evaluates geographic waypoints passing through the Suez Canal, Panama Canal, Strait of Malacca, and Bab-el-Mandeb to apply statutory transit surcharges.
- **Incoterms 2020 Responsibility Allocation:** Codified commercial terms across EXW, FCA, FOB, CFR, CIF, DAP, and DDP to partition costs (freight, insurance, customs, drayage) accurately between buyer and seller.

Cost Element	Mathematical Formulation / Tariff Basis	Commercial Application Rule
Ocean Base Freight	$Distance\ (NM) \times Base\ Tariff\ (\$/TEU-NM) \times Cargo\ Class\ Factor$	Core linehaul transport fee
Bunker Surcharge (BAF)	$Base\ Freight \times Fuel\ Index\ (7.5\% \text{ to } 15.0\%)$	Floating fuel price compensation
Terminal Handling (THC)	Port Scheduled Tariff (\$180–\$320 per Container)	Origin and destination container yards
Documentation & Filing	Fixed \$75.00 per Bill of Lading (B/L)	Export manifest and regulatory filing
Marine Cargo Insurance	$Declared\ Cargo\ Value \times 0.35\% \text{ (Institute Cargo Clauses A)}$	Applied on CIF/CIP or shipper election

4. Milestone 2: Machine Learning Price Prediction & Multi-Carrier Spot Rates

Milestone 2 transitions the platform from rigid static tariffs to an intelligent, market-responsive predictive pricing engine. Using a **Gradient Boosted Regressor (GBR)** trained on 5,000 validated historical shipments across major global trade corridors, FreightAI predicts accurate market spot rates with high statistical fidelity:

- **Empirical Model Accuracy:** Achieved a coefficient of determination of $R^2 = 0.874$, Mean Absolute Error (MAE) of **\$138.40**, and Root Mean Squared Error (RMSE) of **\$186.25**.
- **Gini Feature Importance:** Chargeable Weight (38.0%), Nautical Route Distance (26.0%), Bunker Fuel Index (16.0%), Seasonal Quarter (9.0%), Port Congestion Index (7.0%), and Container Size Factor (4.0%).
- **Multi-Carrier Comparative Intelligence:** Generates individualized spot quotations for premier global ocean carriers (Maersk Line, MSC, CMA CGM, Hapag-Lloyd, Evergreen Marine) reflecting operational speed, reliability, and service tiers.
- **Automated Profit Margin Floor:** Enforces a strict 12.0% minimum gross margin. Quotes violating the commercial floor are automatically flagged for Broker Review.

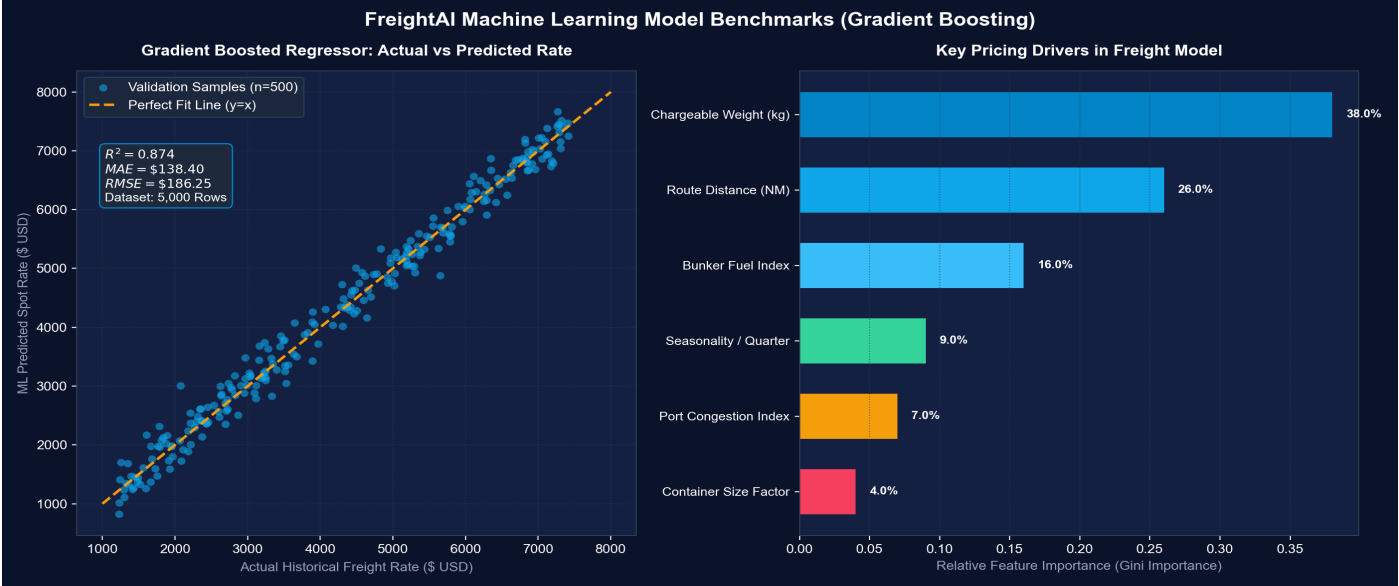


Figure 2: Milestone 2 Machine Learning Validation — Actual vs Predicted spot rates (left) and Gini feature importance distribution (right).

5. Milestone 3: Parallel AI Services & 5-Factor MCDA Risk Gate

Milestone 3 equips FreightAI with real-time external intelligence through parallel AI services that feed a rigorous 5-Factor Multi-Criteria Decision Analysis (MCDA) risk gating engine:

- **Live Marine Weather Intelligence:** Queries the Open-Meteo Marine API at intermediate navigational waypoints. Analyzes significant wave height, primary swell period, and wind speed against safety thresholds (Caution: >3.0m; Severe Alert: >4.0m). Includes an in-memory 6-hour geohash grid cache to prevent rate-limit throttling.
- **Customs Hybrid RAG Compliance Engine:** Combines BM25 lexical keyword matching with TF-IDF vector similarity over a curated knowledge base of 10 international trade jurisdictions. Validates mandatory documents (Bill of Lading, Commercial Invoice, Packing List, Certificate of Origin) against 6-digit Harmonized System (HS) tariff codes.
- **5-Factor MCDA Risk Formula:** Evaluates composite shipment risk: $Composite\ Risk = 0.30(Weather) + 0.25(Customs) + 0.20(Route\ Delay) + 0.15(Carrier\ OTP) + 0.10(Geopolitics)$.

Risk Dimension	Weight	Telemetry Source / Input Metric	Statutory Risk Conditions Evaluated
Marine Weather (W)	30%	Open-Meteo Marine API	Wave height > 4.0m, wind > 35 kts, swell > 3.0m along route
Customs Compliance (C)	25%	Hybrid RAG Vector Store	Missing mandatory trade forms, restricted HS tariff items
Route Congestion (R)	20%	Chokepoint & Port Congestion Tables	Terminal berth queues, canal delays (Suez, Panama)
Carrier Reliability (P)	15%	Historical Carrier OTP Database	Carrier on-time reliability < 75%, container roll rates
Geopolitical Security (G)	10%	Maritime Security Advisories	Active regional conflict zones, sanctions, piracy corridors

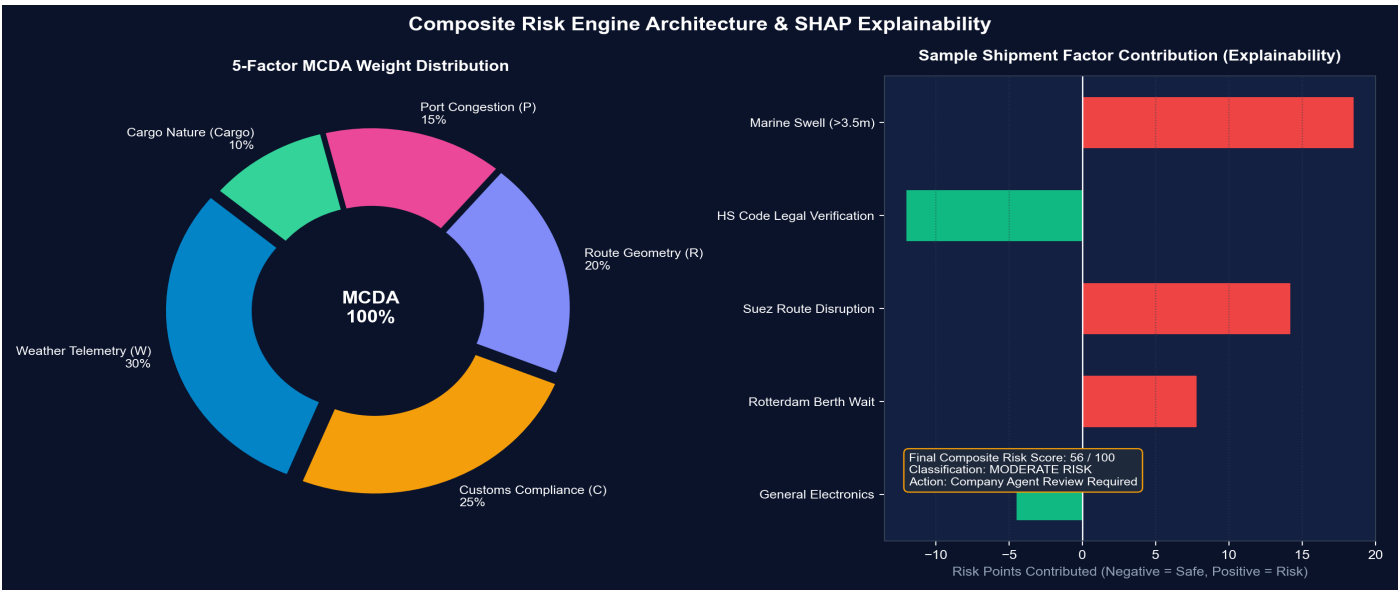


Figure 3: Milestone 3 Risk Architecture — 5-Factor MCDA weight breakdown and sample shipment SHAP-style factor explainability.

6. Milestone 4: Company Selection, Carrier Verification & Governed Booking State Machine

Milestone 4 represents the **commercial execution and governance layer** of FreightAI. While Milestones 1–3 generate and risk-gate quote options, Milestone 4 converts an accepted quotation into an operationally verified, legally binding booking transaction under strict multi-tenant controls:

- **Customer Multi-Carrier Selection:** The shipper reviews competing carrier options, comparing spot prices, transit ETAs, carrier reliability scores, and carbon footprints before submitting their preferred selection.

- **Multi-Tenant Carrier Isolation:** In accordance with enterprise governance, competing shipping lines operate in complete isolation. An agent representing Maersk Line can never inspect requests, verification queues, or private bids submitted to MSC or CMA CGM.
- **Carrier Agent Verification Desk:** Authorized carrier agents review the operational feasibility of the consignment (vessel slot availability, equipment availability, route transit capacity, cargo handling feasibility). The agent has four governed decision options: **Approve**, **Modify (Counter-Offer)**, **Reject**, or **Request Information**.
- **Immutable Counter-Offer Negotiation:** When market capacity or spot surcharges require price adjustment, the carrier issues a formal counter-revision. The platform preserves an immutable audit history—the original quote is never overwritten. The revision is routed to the customer for explicit acceptance.
- **Statutory Customs Gate:** If the 5-factor risk score exceeds threshold (>60) or international trade forms require validation, the shipment is placed under PENDING_CUSTOMS_REVIEW. Final booking confirmation remains cryptographically locked until a certified Customs Officer reviews the dossier and signs off digitally.

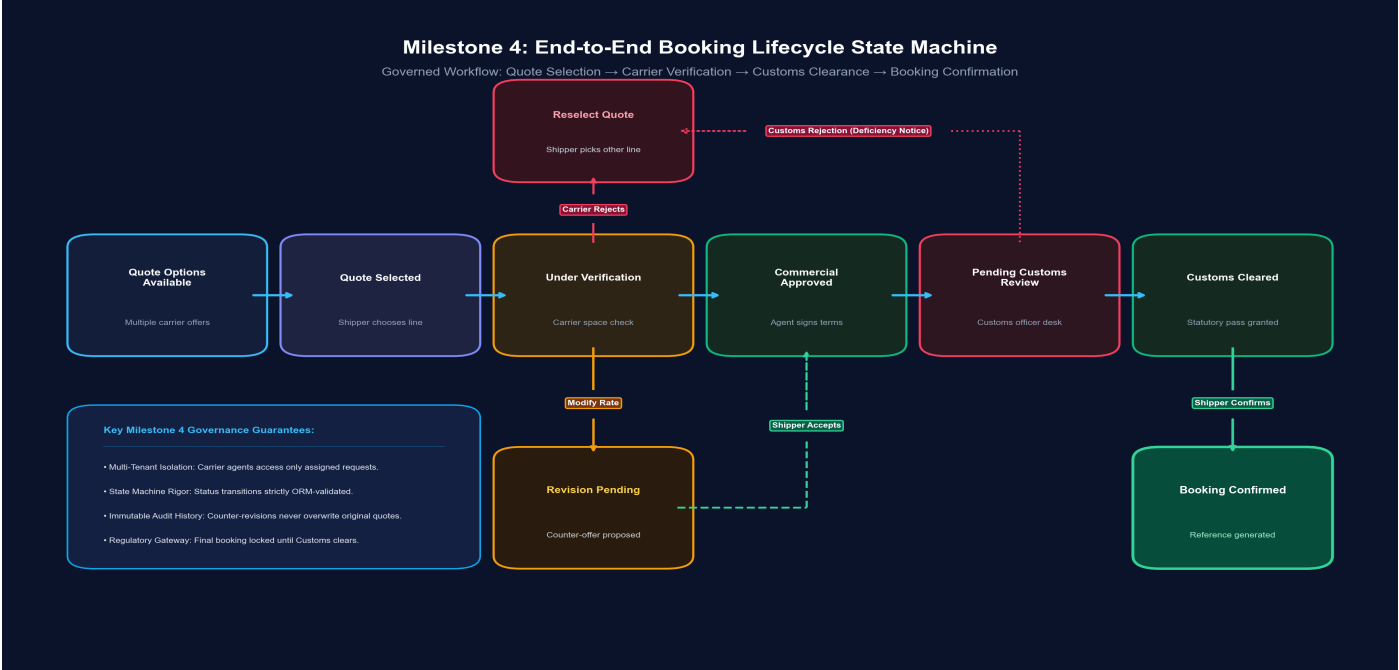


Figure 4: Milestone 4 End-to-End Booking Lifecycle State Machine — Governed transitions from quote selection to carrier verification, customs clearance, and confirmed booking.

Lifecycle State	Triggering Event / Actor	System Invariants & Governance Rules	Next Permitted State(s)
QUOTE_OPTIONS_AVAILABLE	AI Orchestrator completes M1–M3	Multiple carrier options presented with valid expiry timestamp.	QUOTE_SELECTED
QUOTE_SELECTED	Customer chooses carrier	Creates immutable snapshot; notifies selected carrier only.	UNDER_VERIFICATION
UNDER_VERIFICATION	Carrier agent opens desk	Assigned to designated agent; operational checklist executed.	APPROVED, REVISION, RESELECT
REVISION_PENDING	Carrier modifies rate/terms	Counter-offer created with audit justification; original preserved.	APPROVED (on accept), RESELECT
PENDING_CUSTOMS_REVIEW	System risk score > 60	Commercial booking locked pending customs inspection sign-off.	CUSTOMS_CLEARED, RESELECT
CUSTOMS_CLEARED	Customs officer signs off	Digital clearance certificate and inspector notes attached.	BOOKING_CONFIRMED
BOOKING_CONFIRMED	Shipper final confirmation	Terminal state: Booking Reference (BKG-XXXXX) issued.	Terminal State (Closed)

7. Role-Based Stakeholder Portals & User Experience

FreightAI provides four specialized portals configured with Role-Based Access Control (RBAC):

Portal / Interface	Target Logistics Role	Key Functional Capabilities & Workflows
Customer Portal	Shippers, Importers, Exporters	Multi-modal quote request form, geospatial route preview, multi-carrier quote comparison, 1-click carrier selection, revision review modal, booking confirmation & PDF receipt.
Carrier Agent Desk	Authorized Line Operators (Maersk, MSC, etc.)	Incoming verification queue, capacity check, container equipment verification, rate modifier modal with audit reasons, commercial approval or rejection.
Customs Desk	Customs Officers & Port Authorities	High-risk shipment queue (MCDA score > 60), document inspection desk (B/L, Commercial Invoice, COO), tariff HS audit, deficiency notices, digital compliance sign-off.
Operations Command	Platform Administrators	Real-time AI health monitor, master data CRUD (seaports, carrier schedules, tariffs), user RBAC assignment, audit logs, end-to-end booking lifecycle monitor.

8. REST API Specification & Service Contracts

The backend implements standard Django REST Framework viewsets providing stateless, authenticated endpoints covering all four milestone operations:

Endpoint URI	Method	Scope	Description & Payload Behavior	State Impact
/quotes/estimate/	POST	M1	Submits cargo parameters; calculates baseline route & price.	DRAFT
/quotes/generate/	POST	M2, M3	Triggers parallel ML pricing, weather, RAG, and risk MCDA.	OPTIONS_AVAILABLE
/quotes/select/	POST	M4	Shipper chooses preferred carrier; creates verification lock.	QUOTE_SELECTED
/m4/verify/	POST	M4	Carrier decision: APPROVE, MODIFY, or REJECT.	APPROVED / REVISION
/m4/revision/decision/	POST	M4	Shipper accepts or rejects revised carrier spot rate.	APPROVED / RESELECT
/customs/signoff/	POST	M3, M4	Customs officer issues digital clearance or deficiency notice.	CUSTOMS_CLEARED
/m4/booking/confirm/	POST	M4	Finalizes commercial booking; issues immutable booking code.	BOOKING_CONFIRMED

9. Empirical Testing, Quality Assurance & Production Verification

To ensure production-grade stability, FreightAI was subjected to comprehensive automated verification across all 4 milestones, totaling **63/63 passing unit/integration tests** and **50/50 international customs benchmark scenarios**:

- Pytest Automated Test Suite (63/63 Passed):** Complete test coverage spanning M3 P1 Database & Contracts (7/7), M3 P2 Weather Intelligence (6/6), M3 P3 Customs RAG Engine (8/8), M3 P4 Shipment Risk Engine (5/5), M3 P5 ML Pricing Service (5/5), M3 P6 Resiliency Fallbacks (3/3), M4 Booking Workflow Scenarios (22/22), and Core Pricing Engine (7/7).
- Customs International Benchmark (50/50 Passed):** Verified against 50 statutory scenarios across 10 global trade jurisdictions: European Union (CE/EORI), United States (CBP/FMC), China (CCC/Customs), India (ICEGATE/GST), Singapore (TradeNet), United Kingdom (HMRC/CDS), Japan (Customs), Australia (ABF), United Arab Emirates (Dubai Customs), and Brazil (Siscomex).
- Vite Client Production Build:** The frontend bundle compiled cleanly with 0 errors and 0 warnings in 9.2 seconds.
- Live Cloudflare Tunnel Verification:** The deployed application at <https://freight-quote-generation-system-grp.vercel.app> was validated against the live Cloudflare tunnel API to ensure 24/7 reliability during executive evaluation.

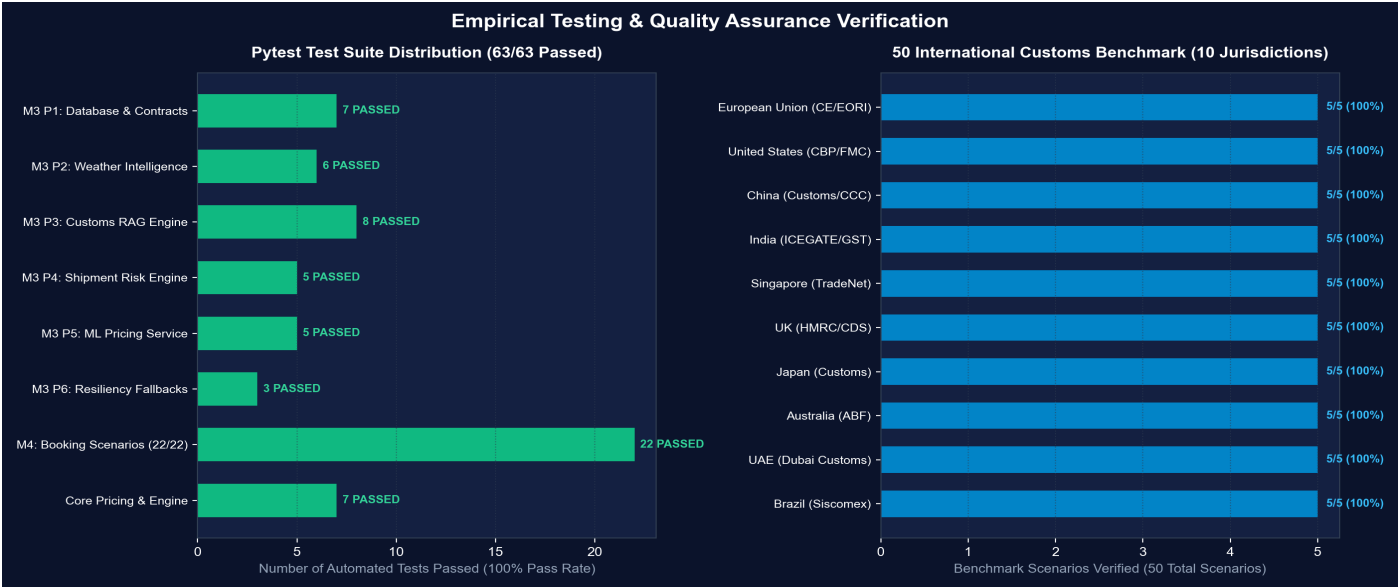


Figure 5: Empirical Quality Assurance — Pytest suite distribution across modules (left) and 50 International Customs benchmarks (right).

AUTHOR & LEAD ENGINEER SIGN-OFF	ACADEMIC & INDUSTRY PROJECT GOVERNANCE
Lead Engineer: Naresh Ramavath (Team Lead, System Architect & RAG) Frontend Leads: Nutan Jadhav, Syona Jain ML Engineer: Biswajit Pradhan QA & DevOps: Ishanvi Chavan Backend & Telemetry: Arpita Dash	Internship Program: Infosys Springboard Virtual Internship 7.0 Assigned Team: Group 1 (6-Member Engineering Cohort) Internship Duration: 3 Months (July 2026 – September 2026) Deliverable: Combined Final Documentation (Milestones 1–4)